

THEORETICAL MODEL

$$\text{Basket} = \beta_0 + \beta_1 \cdot \text{Income} + \varepsilon$$

ESTIMATED MODEL

$$\hat{\text{Basket}}_i = 3.16 + 0.1097 \cdot \text{Income}_i$$

Call:

```
lm(formula = basket ~ income, data = Basket)
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Residuals:

Min	1Q	Median	3Q	Max
-22.0429	-7.2305	0.0133	6.3991	23.6122

Five number summary of residuals

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.1638282	3.9037148	0.81	0.42
income	0.10109748	0.0006467	16.97	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.793 on 97 degrees of freedom

Multiple R-squared: 0.748, Adjusted R-squared: 0.7454

F-statistic: 288 on 1 and 97 DF, p-value: < 2.2e-16

R^2
(coefficient of determination)

P-value for the hypothesis test

$$H_0: \beta_1 = 0$$

$$H_1: \beta_1 \neq 0$$

This p-value is calculated from t-score = 16.97

(which comes from a t_{n-2} distribution under H_0)

If the p-value is below the significance level α we REJECT H_0 and we say that the variable INCOME IS SIGNIFICANT

If the p-value is above the significance level α we DO NOT REJECT H_0 and we say that the variable INCOME IS NOT SIGNIFICANT

$$H_0: \beta_0 = 0$$

$$H_1: \beta_0 \neq 0$$

The p-value is 0.42

DO NOT REJECT H_0

INTERCEPT IS

NOT SIGNIFICANT

multiple linear regression stuff